NBA: Luxury Tax, Super Teams, and the fairness of the system

Pedro Lopes
Diana Carvalho
Tara Schubert
Matteo Paonessa
Irene Sforza

Instituto Português de Administração de Marketing
Marketing Research Methods
Professor Ronnie Figueiredo

Table of Contents

Introduction **3**
Background 3
Research Question 3
Objectives 4
Methodology 5
Data Analysis (descriptive) quantitative **7**
Data Sources 7
Data Collection and Preprocessing 7
Hypothesis (based on data we collected in part 3) **8**
Hypotheses generation (H1, H2, H3) 8
Results and Data Visualization 12
Interpretation **15**
Season 2013/2014 17
Season 2014/2015 17
Season 2015/2016 18
Season 2016/2017 18
Season 2017/2018 19
Season 2018/2019 19
Season 2019/2020 20
Season 2020/2021 20
Season 2021/2022 21
Season 2022/2023 22
Relevant teams' analysis 24
Regression Model 27
Conclusions **28**
Summary of Findings 28
Limitations 29
Indirect observation 29
Focus Groups 30
Question analysis 31
In-depth interview 32
Future Application and Recommendations 33
Appendices **34**
References 35

Introduction

Background

The National Basketball Association (NBA) is the national body organised to operate a league consisting of professional basketball teams (Grossman, N. (2014) currently consisting of 30 teams: 29 in the United States and 1 in Canada, divided into an Eastern and a Western Conference, which in turn are each divided into three divisions. (Cf. Grossman, N. (2014)

The organisational structure of the NBA is diverse. First, the relationship between members of the NBA is contractual, as the league's constitution describes the NBA as a "contract among the Members".

As in any other sporting competition, one of the topics in the spotlight is the rivalry among existing competitors. In contrast to most other industries, professional sports leagues require a certain degree of collaboration between competitors. This phenomenon is called the "Louis – Schmeling - Paradox" because you cannot play basketball or any team sport without having an opponent (Cf. Congdon-Hohman, J., Matheson, V. (2022). That is another reason why so-called "super teams" are dangerous to the league, and a competitive balance is essential. Theoretically, the NBA consists of 30 equally balanced competitors to maintain the game's suspense and attract fans.

That means that the distribution of talent must also be fair and equal, but that is not always the case because of the financial disadvantages.

Some of the main methods to ensure the competitive balance of the league are the salary cap and the luxury tax system: NBA has a soft salary cap, meaning that franchises can exceed it, but, because of that, the league has also introduced a penalty system with the Luxury Tax.

The problem is that some teams are willing to pay the luxury tax, because most of the time if they exceed the salary cap, they can sign better players with higher salary contracts and have a better chance of winning the Championship.

Research Question

How does the luxury tax create an unfair competitive system in the NBA?

Objectives

The focus of our study is to understand whether and how the luxury tax creates an unfair environment for competition. To do this, we used two variables: the team's revenue and the team's salaries, which, as we hypothesised, influence an additional variable, the win rate.

Why? Because teams with more revenue (and thus more money from merchandising or winnings) are the ones that tend to invest more in players (thus giving higher salaries) and we consequently assumed to have a higher win rate.

We therefore focus on how the NBA's competitive environment works in order to understand whether this problem exists through three different constructs and, if so, to find alternative solutions if our assumptions are confirmed.

Methodology

After analyzing the data and reviewing the literature, the variables to be taken into consideration are the number of wins (win rate), the revenue of each team, and the team's salaries. In this case, the team's revenue and salaries are our independent variables because, as hypothesized, they would be the ones that, when changing, would influence the change in the dependent variable, namely the win rate.

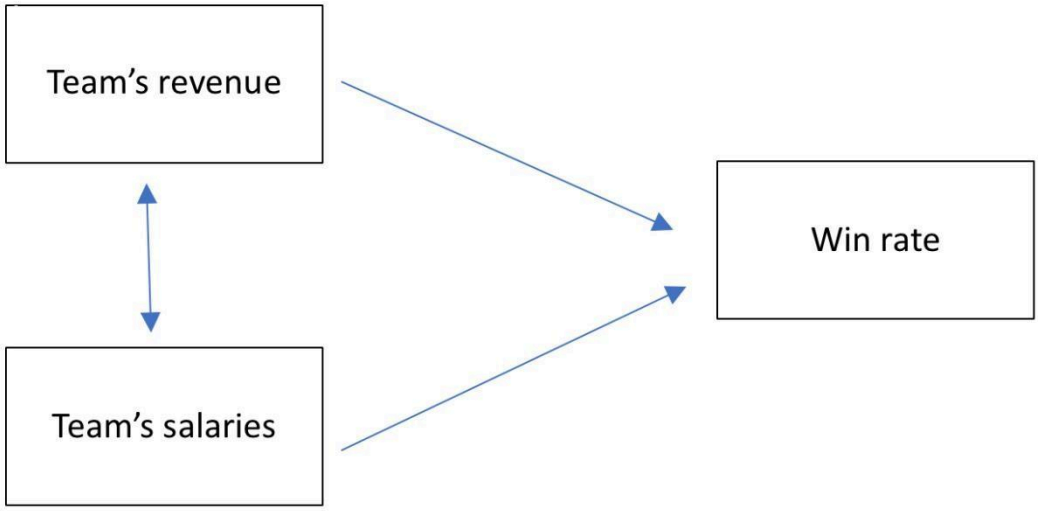


Figure 1 – Methodology Diagram

V1: TEAM'S REVENUE	• Simmons, R., Berri, D.J. (2011), "Mixing the princes and the paupers: Pay and performance in the National Basketball Association," Labour Economics Vol. 18, pp. 381-388. • Berri, D.J., Jewell, R.T. (2004), "Wage inequality and firm performance: professional basketball's natural experiment," Atlantic Economic Journal Vol. 32, pp. 130-139.
V2: TEAM'S SALARIES	• Tao, Y., Chuang, H., & Lin, E. S. (2016). Compensation and performance in Major League Baseball: Evidence from salary dispersion and team performance. *International Review of Economics & Finance*, *43*, 151–159. https://doi.org/10.1016/j.iref.2015.10.037 • *Coates, D., Frick, B., Jewell, T. (2014), "Superstar Salaries and Soccer Success: The Impact of Designated Players in Major League Soccer," Journal of Sports Economics, first published on August 21, 2014 doi: 10.1177/1527002514547297.* • *Mathur, L., Mathur, I., Rangan, N. (1997), "The wealth effects associated with a celebrity endorser: The Michael Jordan phenomenon," Journal of Advertising Research, Vol. 37, No. 3, pp. 67–73.*
V3: WIN RATE	• Shorin, G. (2017). Team Payroll Versus Performance in Professional Sports: Is Increased Spending Associated with Greater Success? Retrieved 2023, from https://sites.duke.edu/djepapers/files/2017/06/grantshorin-dje.pdf • Cyrenne, P. (2017). SALARY INEQUALITY, TEAM SUCCESS, LEAGUE POLICIES, AND THE SUPERSTAR EFFECT. *Contemporary Economic Policy*, *36*(1), 200–214. https://doi.org/10.1111/coep.12217

Table 1 – Methodology Table

Data Analysis (descriptive) quantitative

Data Sources

Given the nature of the research, we exclusively used secondary data. This data was sourced from official platforms, such as the NBA website and ESPN's statistics portal, to address our research question. The data encompasses the performance of the 30 NBA teams across five years (season 2018/19-2022/23). Our analysis will focus on each team's win rate percentage, combined player salaries, and the team's revenues.

Data Collection and Preprocessing

By opting for the play season 2022/2023, we aimed to capture the most recent and relevant insights into the performance and financial aspects of each NBA team. The inclusion of all 30 teams in our dataset ensures a holistic view of the league, allowing for comprehensive analysis and meaningful conclusions.

To facilitate efficient analysis we used Excel. On the sheet, numerical values are rounded, as a common practice to enhance precision and accommodate the spreadsheet's functionality. While the rounding of numbers may impact the exact precision, it serves the practical purpose of creating accurate computations within the chosen analytical tool. Our commitment to methodological rigor and data integrity underscores the robustness of our approach in examining the dynamics of the NBA landscape during the specified seasons.

Once each of the three datasets was ready, we started our analysis in Excel using graphic lines and crossing the values to create a coherent and clear data visualization.

Hypothesis (based on data we collected in part 3)

Hypotheses generation (H1, H2, H3)

According to Statista, during the 2022/23 season, the average revenue per franchise in the National Basketball Association (NBA) stood at 353 million U.S. dollars. Reading through an article by P. Menon (2016), its content outlines several key points, it seems in fact the teams that have more revenues are the ones that pay more luxury tax, and this is associated with a boost in team performance in terms of wins.

Let's take an example that we found in an article in Forbes[1]: in the 2021-22 season, **the Warriors generated the most revenue** ($765 million after paying their revenue-sharing check) and the most operating income ($206 million) **in NBA history** as they won their fourth title in eight years and played their first full season with fans in the new Chase Center. According to Statista, in 2023, the franchise value of the Golden State Warriors reached 7.7 billion U.S. dollars (as we already highlighted). This was an increase of 10% from the previous year when the figure stood at 7 billion U.S. dollars for the National Basketball Association franchise.

According to the data of DraftKings Sportsbook, the Warriors have for this new season the fifth-best odds to win the 2023 NBA Finals at (+950). Ahead of them are the Milwaukee Bucks (+265), Boston Celtics (+320), Phoenix Suns (+425), and Philadelphia 76ers (+900). **So, they have generated the best revenue over the years and are now one of the most likable teams to win.**

Maybe the luxury tax might not be punitive enough: this could imply that the fine itself is insufficient or that the overall cost of the tax is less than the billed rate of the fine. Theoretically, **stronger regular and postseason performance, as well as the presence of a superstar player, should boost a team's revenues and winning possibilities**.

It is suggested that a **profit-maximizing owner has a logical reason to exceed the luxury threshold**, as this not only increases the team's chances of success in the playoffs but also leads to incremental playoff revenue, increased ticket sales, and higher prices in the following year.

However, a potential challenge arises when considering the financial implications of the luxury tax, especially for lower revenue teams. **Indeed, the luxury tax is regressive, disproportionately affecting smaller teams either financially or performance-wise**. The

higher spending by larger market teams, despite facing the luxury tax, is seen as a nuisance rather than a significant impediment.

In summary, the article contends that **the luxury tax**, instead of promoting competitive balance, **may contribute to a widening gap between larger and smaller market teams**, ultimately questioning the necessity and effectiveness of the luxury tax in the context of team revenues and performance.

This process made us elaborate the first alternative hypothesis:

H1: Revenues of teams are associated with the teams' win rate.

When analysing teams with higher incomes we can detect a higher expenditure on salaries. With this, the team can sign better players, creating the ultimate roster, completed with the season's greatest players, which, while not meaning that the team will have guaranteed victories, does increase its winning odds.

Therefore, some teams are willing to go over the salary cap imposed by the NBA, because it will most likely result in profitable wins. **According to Forbes, not only are these teams 30% more likely to make it to the playoffs, but they also continue to outperform non-taxpayer teams throughout them.**

Some proof of this association comes from the Warriors case: their salary cap was $112.4 million, but the payroll of the Warriors was over $184 million, causing them to exceed the cap by more than $70 million, resulting in the franchise paying $170.3 million in luxury taxes, which is also the NBA record. But by exceeding the salary cap, the Warriors also won four championships in eight seasons, confirming this approach's success.

Even if the correlation between dollars spent and the number of regular season wins doesn't rebound, the NBA has returned to a dynamic where the **top spenders routinely dominate playoff berths and bottom spenders rarely qualify.** Accordingly, luxury tax teams are 1.3x as likely to make the playoffs as teams that don't pay the tax (63% v. 49%) – an advantage that has largely held since the tax's inception and across the Eastern and Western

conferences. Only twice since 2002 have tax teams been less likely to make the post-season than non-tax teams.

Prior to the tax's introduction, the correlation between payroll and performance had been increasing steadily. In the years immediately following the tax's introduction, this correlation plummeted.

So based on the literature we have elaborated our second hypothesis:
H2: Team' salaries are associated with the teams' win rate.

As in most pro sports, NBA teams make their money from a combination of ticket sales, suite sales, naming rights and local TV and radio deals, as well as league-wide media, merchandising and sponsorship revenues. But how do these franchises spend their revenue?

The teams who have higher revenues will be able to increase their salary expenditures. As supported by the data, these teams are paying higher salaries and consequently going over the salary cap.

For the luxury tax to be effective, team owners must truly feel the cost of every salary dollar spent above the threshold. Faced with such significant and persistent annual gains in equity value, the owners of many major teams likely view the luxury tax as an inevitable cost of doing business, or simply the investment required to win a title. Rationally, of course, a dollar spent is a dollar spent – but to owners, each year has been more valuable than the last, which means the net effect of the tax may go unfelt.

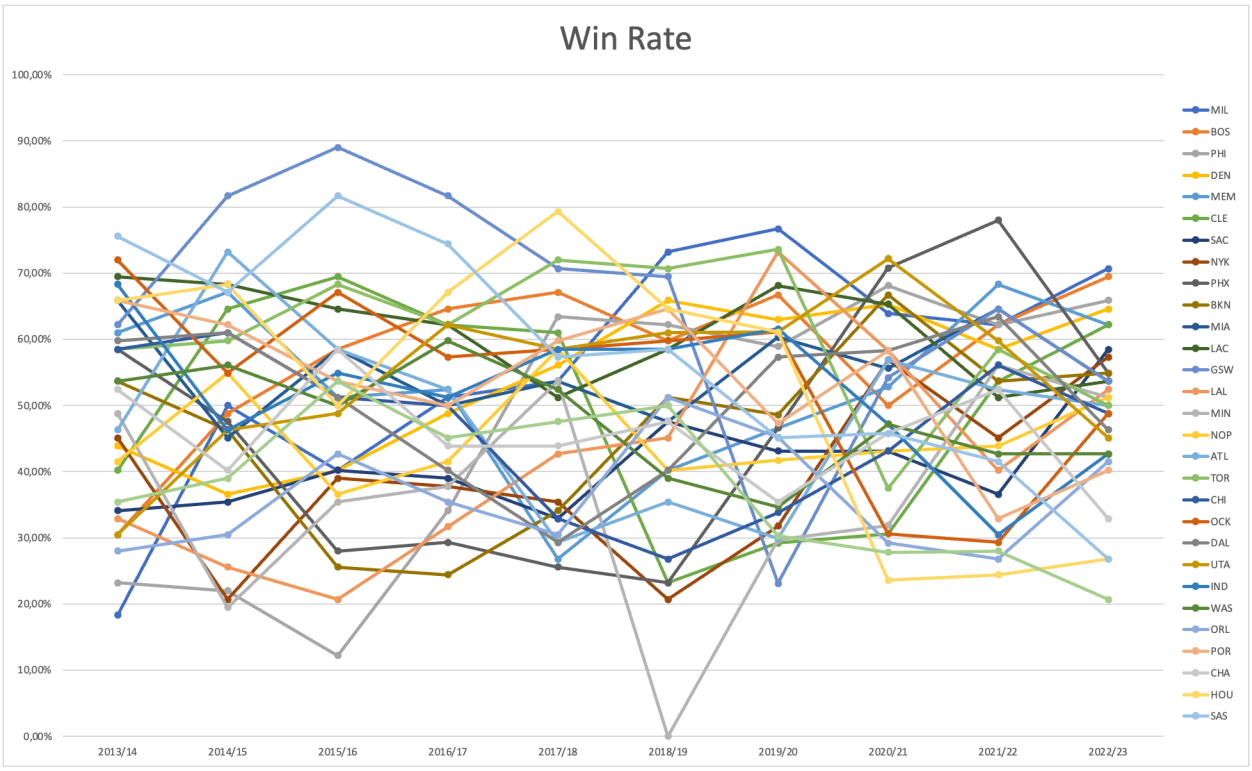
The Brooklyn Nets are a perfect example. One year after the Nets' billionaire ex-owner, Mikhail Prokhorov incurred the $86M luxury tax fine (which helped drive operating income down from -$19M to -$99M year-over-year) **Forbes doubled its estimate of the team's value to $1.5B, and Prokhorov put the team up for sale for a similar amount** (the precise price is unknown given the team is bundled with other assets, including the Barclays Center).

This brought us to our final hypothesis:
H3: Revenues of teams are associated with the teams' salaries.

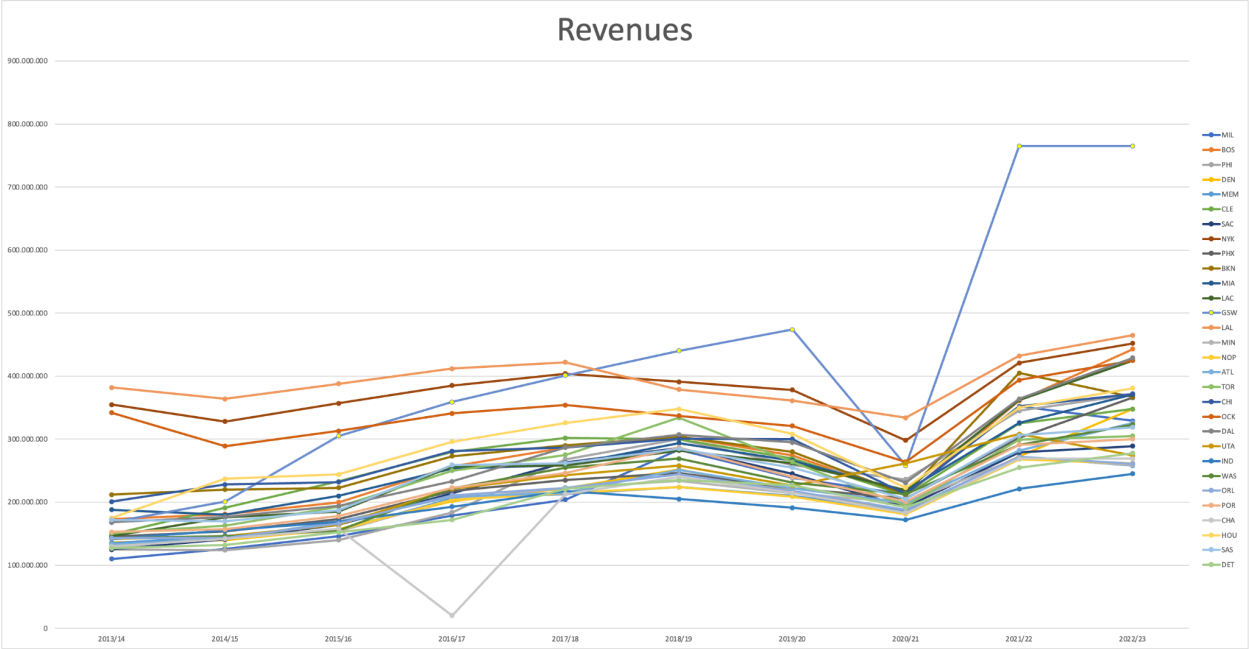
The null hypothesis is therefore based on the disconfirmation of the expectations arising from the literature review.

H0: Revenues of teams, salaries of teams, and win rate are not associated.

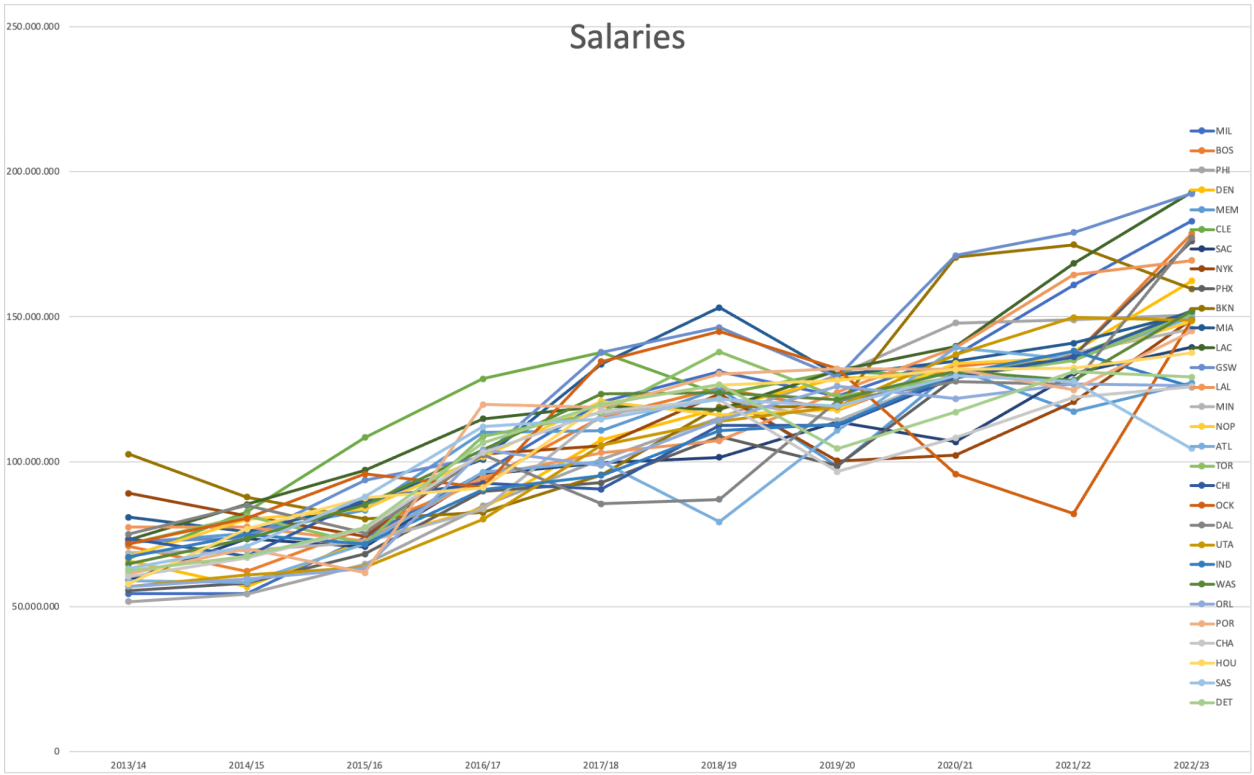
Results and Data Visualization



Analysing the graph of the win rate of the 30 teams over a period of ten seasons does not offer many conclusions. For this reason, we decided to take a closer look at what we consider to be the most interesting teams, for example those that have shown particularly high or low values or major development. This also applies to the following two graphs. However, it can be said that the win rate has become more balanced in recent years. The teams are all very close to each other and there are hardly any major peaks as it was the case in previous seasons. This suggests that the league as a whole has become more balanced.

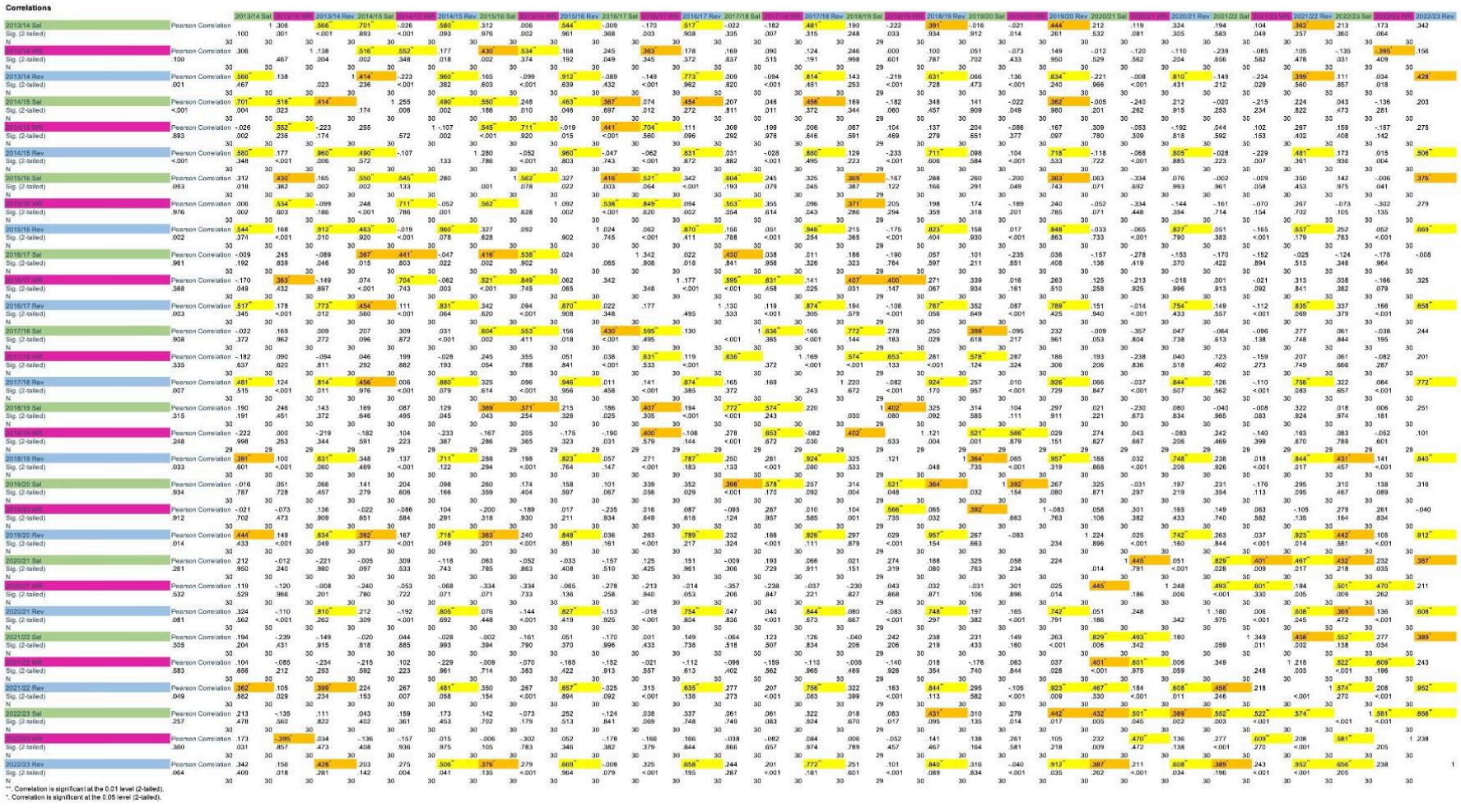


When it comes to revenue, the biggest noticeable thing is definitely the sharp drop in revenue in the 2020/21 season due to the COVID-19 pandemic. Since the following season, however, the trend has been strongly positive again. The graph also shows that the teams' revenues are not significantly interrelated, which may be due to the popularity, the location of the team and other different factors. This means that the teams with higher earnings will probably always be at the top and the same applies to the teams at the bottom. However, it can also be seen that the distance between the top earning teams and the rest has decreased over the last few seasons. Furthermore, it can be seen that the revenue curve is relatively the same for all teams except for two teams: the Golden State Warriors and the Charlotte Hornets.



Regarding salaries, we can also see an overall significant increase in over the last ten years. In general, it can also be said that the salaries of the different teams now differ a lot more than they did 10 years ago. However, there are also teams that have cut their salaries over the years, which can be due, for example, trading expensive players for draft picks or completely rebuilding a team.

Interpretation



Correlations

		2013/14 Sal	2013/14 WR	2013/14 Rev	2014/15 Sal	2014/15 WR	2014/15 Rev	2015/16 Sal	2015/16 WR	2015/16 Rev	2016/17 Sal	2016/17 WR	2016/17 Rev	2017/18 Sal	2017/18 WR	2017/18 Rev	2018/19 Sal	2018/19 WR	2018/19 Rev	2019/20 Sal	2019/20 WR	2019/20 Rev	2020/21 Sal	2020/21 WR	2020/21 Rev	2021/22 Sal	2021/22 WR	2021/22 Rev	2022/23 Sal	2022/23 WR	2022/23 Rev
2013/14 Sal	Pearson Correlation	1	.306	.566**	.701**	-.026	.580**	.312	.006	.544**	-.009	-.170	.517**	-.022	-.182	.481**	.190	-.222	.391*	-.016	-.021	.444*	.212	.119	.324	.194	.104	.362*	.213	.173	.342
	Sig. (2-tailed)		.100	.001	<.001	.893	<.001	.093	.976	.002	.961	.368	.003	.908	.335	.007	.315	.248	.033	.934	.912	.014	.261	.532	.081	.305	.583	.049	.257	.360	.064	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2013/14 WR	Pearson Correlation	.306	1	.138	.516**	.552**	.177	.430*	.534**	.168	.245	.363*	.178	.169	.090	.124	.246	.000	.100	.051	-.073	.149	-.012	-.120	-.110	-.239	-.085	.105	-.135	.395*	.166	
	Sig. (2-tailed)	.100		.467	.004	.002	.348	.018	.002	.374	.192	.049	.345	.372	.637	.515	.191	.998	.601	.787	.702	.433	.950	.529	.562	.204	.656	.582	.478	.031	.409	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2013/14 Rev	Pearson Correlation	.566**	.138	1	.414*	.223	.960**	.165	-.099	.912**	-.089	-.149	.773**	.009	-.094	.814**	.143	-.219	.631**	.066	.136	.634**	-.221	-.008	.810**	-.149	-.234	.399*	.111	.034	.428*	
	Sig. (2-tailed)	.001	.467		.023	.236	<.001	.382	.603	<.001	.639	.432	<.001	.962	.620	<.001	.451	.253	<.001	.728	.473	<.001	.240	.966	<.001	.431	.212	.029	.560	.857	.018	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2014/15 Sal	Pearson Correlation	.701**	.516**	.414*	1	.255	.490**	.550**	.248	.463**	.367*	.074	.454*	.207	.048	.456*	.169	-.182	.348	.141	-.022	.562**	-.005	-.240	.212	-.020	-.215	.224	.043	-.136	.203	
	Sig. (2-tailed)	<.001	.004	.023		.174	.006	.002	.186	.010	.046	.697	.012	.272	.811	.011	.372	.344	.060	.457	.909	.049	.980	.201	.262	.915	.253	.234	.822	.473	.281	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2014/15 WR	Pearson Correlation	.026	.552**	.223	.255	1	.107	.545**	.711**	.019	.441*	.704**	.111	.309	.199	.006	.087	.104	.137	.204	.086	.167	.309	.053	.192	.044	.102	.287	.159	.157	.275	
	Sig. (2-tailed)	.893	.002	.236	.174		.572	.002	<.001	.920	.015	<.001	.560	.096	.292	.978	.646	.591	.469	.279	.651	.377	.097	.780	.309	.818	.592	.153	.402	.408	.142	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2014/15 Rev	Pearson Correlation	.580**	.177	.960**	.490**	.107	1	.280	-.052	.960**	-.047	-.062	.831**	.031	-.028	.880**	.129	-.233	.711**	.098	.104	.718**	-.118	-.068	.805**	-.028	-.229	.481**	.173	.015	.506**	
	Sig. (2-tailed)	<.001	.348	<.001	.006	.572		.133	.786	<.001	.803	.743	<.001	.872	.882	<.001	.495	.223	<.001	.606	.584	<.001	.533	.722	<.001	.885	.223	.007	.361	.936	.004	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2015/16 Sal	Pearson Correlation	.312	.430*	.165	.550**	.545**	.280	1	.562**	.327	.416*	.521**	.342	.604**	.245	.325	.369*	-.167	.288	.260	-.200	.363*	.063	.334	.076	-.002	-.009	.350	.142	-.006	.376*	
	Sig. (2-tailed)	.093	.018	.382	.002	.002	.133		.001	.078	.022	.003	.064	<.001	.193	.079	.045	.387	.122	.166	.291	.049	.743	.071	.692	.993	.961	.058	.453	.975	.041	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2015/16 WR	Pearson Correlation	.006	.534**	-.099	.248	.711**	-.052	.562**	1	.092	.538**	.849**	.094	.553**	.355	.096	.371*	.205	.198	.174	-.189	.240	-.052	-.334	-.144	-.161	-.070	.267	-.073	-.302	.279	
	Sig. (2-tailed)	.976	.002	.603	.186	<.001	.786	.001		.628	.002	<.001	.620	.002	.054	.614	.043	.286	.294	.359	.318	.201	.785	.071	.448	.394	.714	.154	.702	.705	.135	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2015/16 Rev	Pearson Correlation	.544**	.168	.912**	.463**	.019	.960**	.327	.092	1	.024	.062	.870**	.156	.051	.946**	.215	-.175	.823**	.158	.017	.848**	-.033	-.065	.827**	.051	-.165	.857**	.252	.052	.669**	
	Sig. (2-tailed)	.002	.374	<.001	.010	.920	<.001	.078	.628		.902	.745	<.001	.411	.788	<.001	.254	.365	<.001	.404	.930	<.001	.863	.733	<.001	.790	.383	<.001	.179	.783	<.001	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2016/17 Sal	Pearson Correlation	-.009	.245	-.089	.367*	.441*	-.047	.416*	.538**	.024	1	.342	.022	.430*	.038	.011	.186	-.190	.057	.101	-.235	.036	-.157	-.278	-.153	-.170	-.152	-.025	-.124	-.176	-.008	
	Sig. (2-tailed)	.961	.192	.639	.046	.015	.803	.022	.002	.902		.065	.908	.018	.841	.956	.326	.323	.764	.597	.211	.851	.408	.136	.419	.370	.422	.894	.513	.348	.964	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2016/17 WR	Pearson Correlation	-.170	.363*	-.149	.074	.704**	-.062	.521**	.849**	.062	.342	1	.177	.595**	.831**	.141	.407*	.400*	.271	.339	.016	.263	.125	-.213	-.018	.001	-.021	.313	.038	-.166	.325	
	Sig. (2-tailed)	.368	.049	.432	.697	<.001	.743	.003	<.001	.745	.065		.348	<.001	<.001	.458	.025	.031	.147	.067	.934	.161	.510	.258	.925	.996	.913	.092	.841	.382	.079	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2016/17 Rev	Pearson Correlation	.517**	.178	.773**	.454*	.111	.831**	.342	.094	.870**	.022	.177	1	.130	1.000**	.119	.874**	.165	.787**	.352	.087	.789**	.151	-.014	.754**	.149	-.112	.835**	.337	.166	.658**	
	Sig. (2-tailed)	.003	.345	<.001	.012	.560	<.001	.064	.620	<.001	.908	.348		.495	.533	<.001	.305	.379	<.001	.056	.849	<.001	.425	.940	<.001	.433	.557	<.001	.069	.378	<.001	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2017/18 Sal	Pearson Correlation	-.022	.169	.009	.207	.309	.031	.604**	.553**	.156	.430*	.595**	.130	1	.636**	.165	.772**	.278	.250	.398*	-.095	.232	-.009	-.357	.047	-.064	-.096	.277	.061	-.038	.244	
	Sig. (2-tailed)	.908	.372	.962	.272	.096	.872	<.001	.002	.411	.018	<.001	.495		<.001	.385	<.001	.144	.183	.029	.618	.217	.961	.053	.804	.738	.613	.138	.748	.844	.195	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2017/18 WR	Pearson Correlation	-.182	.090	-.094	.046	.199	-.028	.245	.355	.051	.038	.831**	.119	.636**	1	.169	.574**	.653**	.281	.578**	.287	.186	.193	-.238	.040	.123	-.159	.207	.061	-.082	.201	
	Sig. (2-tailed)	.335	.637	.620	.811	.292	.882	.193	.054	.788	.841	<.001	.533	<.001		.372	<.001	<.001	.133	<.001	.124	.324	.306	.206	.836	.518	.402	.273	.749	.666	.287	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2017/18 Rev	Pearson Correlation	.481**	.124	.814**	.456*	.006	.880**	.325	.096	.946**	.011	.141	.874**	.165	.169	1.000**	.220	-.082	.924**	.257	.010	.926**	.066	-.037	.844**	.126	-.110	.756**	.322	.084	.772**	
	Sig. (2-tailed)	.007	.515	<.001	.011	.976	<.001	.079	.614	<.001	.956	.458	<.001	.385	.372		.243	.672	<.001	.170	.957	<.001	.729	.847	<.001	.507	.562	<.001	.083	.657	<.001	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2018/19 Sal	Pearson Correlation	.190	.246	.143	.169	.087	.129	.369*	.571**	.215	.186	.407*	.194	.772**	.574**	.220	1	.402*	.325	.314	.104	.297	.021	-.230	.080	-.040	-.008	.322	.018	.006	.251	
	Sig. (2-tailed)	.315	.191	.451	.372	.646	.495	.045	.043	.254	.326	.025	.305	<.001	<.001	.243		.030	.080	.092	.585	.111	.911	.221	.673	.834	.965	.083	.824	.974	.181	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2018/19 WR	Pearson Correlation	-.222	.000	-.219	-.182	.104	-.233	-.167	.205	-.175	-.190	.406*	-.108	.278	.653**	-.082	.402*	1	.121	.521**	.586**	.029	.274	.043	-.083	.242	-.140	.163	.083	-.052	.101	
	Sig. (2-tailed)	.248	.998	.253	.344	.591	.223	.387	.286	.365	.323	.031	.579	.144	<.001	.672	.030		.533	.004	<.001	.879	.151	.827	.667	.206	.469	.399	.670	.789	.601	
	N	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29	29
2018/19 Rev	Pearson Correlation	.391*	.100	.631**	.348	.137	.711**	.288	.198	.823**	.057	.271	.787**	.250	.261	.924**	.325	.121	1	.364*	.065	.957**	.188	.032	.748**	.238	.018	.844**	.431*	.141	.840**	
	Sig. (2-tailed)	.033	.601	<.001	.060	.469	<.001	.122	.294	<.001	.764	.147	<.001	.183	.133	<.001	.080	.533		.048	.735	<.001	.319	.868	<.001	.206	.926	<.001	.017	.457	<.001	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2019/20 Sal	Pearson Correlation	-.016	.051	.066	.141	.204	.098	.260	.174	.158	.101	.339	.352	.398*	.578**	.257	.314	.521**	.364*	1	.392*	.267	.325	-.031	.197	.231	-.176	.295	.310	.138	.316	
	Sig. (2-tailed)	.934	.787	.728	.457	.279	.606	.166	.359	.404	.597	.067	.056	.029	<.001	.170	.092	.004	.048		.032	.154	.080	.871	.297	.219	.354	.113	.095	.467	.089	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2019/20 WR	Pearson Correlation	-.021	-.073	.136	-.022	-.086	.104	-.200	-.189	.017	-.235	.016	.087	-.095	.267	.010	.104	.566**	.065	.392*	1	-.083	.058	.301	.165	.149	.063	-.105	.279	.261	-.040	
	Sig. (2-tailed)	.912	.702	.473	.909	.651	.584	.291	.318	.930	.211	.934	.649	.618	.124	.957	.585	.001	.735	.032		.663	.763	.106	.382	.433	.740	.582	.135	.164	.834	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2019/20 Rev	Pearson Correlation	.444*	.149	.634**	.362*	.167	.718**	.363*	.240	.848**	.036	.263	.789**	.232	.186	.926**	.297	.029	.957**	.267	-.083	1	.224	.025	.742**	.263	.037	.923**	.442*	.105	.912**	
	Sig. (2-tailed)	.014	.433	<.001	.049	.377	<.001	.049	.201	<.001	.851	.161	<.001	.217	.324	<.001	.111	.879	<.001	.154	.663		.234	.896	<.001	.160	.844	<.001	.014	.581	<.001	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2020/21 Sal	Pearson Correlation	.212	-.012	-.221	-.005	.309	-.118	.063	-.052	-.033	-.157	.125	.151	-.009	.193	.066	.021	.274	.188	.325	.058	.224	1	.445*	.051	.829**	.401*	.467**	.432*	.232	.367*	
	Sig. (2-tailed)	.261	.950	.940	.980	.097	.533	.743	.785	.863	.408	.510	.425	.961	.306	.729	.911	.151	.319	.080	.763	.234		.014	.791	<.001	.028	.009	.017	.218	.035	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2020/21 WR	Pearson Correlation	.119	-.120	-.008	-.240	-.053	-.068	-.334	-.334	-.065	-.278	-.213	-.014	-.357	-.238	-.037	-.231	-.031	-.027	-.213	.043	.032	-.031	.301	.025	1.000**	.248	.493**	.601**	.184	.501**	.470**	.211	
	Sig. (2-tailed)	.532	.529	.966	.201	.780	.722	.071	.071	.733	.136	.258	.940	.053	.206	.847	.221	.827	.868	.871	.106	.896	.914		.186	.006	<.001	.330	.005	.009	.262	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2020/21 Rev	Pearson Correlation	.324	-.110	.810**	.212	-.192	.805**	.076	-.144	.827**	-.153	-.018	.754**	.047	.040	.844**	.080	-.083	.748**	.197	.165	.742**	.051	.248	1.000**	.006	.608**	.369*	.136	.608**		
	Sig. (2-tailed)	.081	.562	<.001	.262	.309	<.001	.692	.448	<.001	.419	.925	<.001	.804	.836	<.001	.673	.667	<.001	.297	.382	<.001	.791	.186		.342	.975	<.001	.045	.472	<.001	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2021/22 Sal	Pearson Correlation	.194	-.239	-.149	-.020	.044	-.028	-.002	-.161	.051	-.170	.001	.149	-.064	.123	.126	-.040	.242	.238	.231	.149	.263	.829**	.493**	.180	1.000**	.458*	.552**	.277	.389*	
	Sig. (2-tailed)	.305	.204	.431	.915	.818	.885	.993	.394	.790	.370	.996	.433	.738	.518	.507	.834	.206	.206	.219	.433	.160	<.001	.006	.342		.059	.011	.002	.138	.034	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2021/22 WR	Pearson Correlation	.104	-.085	-.234	-.215	.102	-.229	-.009	-.070	-.165	-.152	-.021	-.112	-.099	-.159	-.110	-.008	-.140	.018	-.176	.063	.037	.401*	.601**	.006	.349	1.000**	.218	.522**	.609**	.243	
	Sig. (2-tailed)	.583	.656	.212	.253	.592	.223	.961	.714	.383	.422	.913	.557	.613	.402	.562	.965	.469	.926	.354	.740	.844	.028	<.001	.975	.059		.246	.003	<.001	.196	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2021/22 Rev	Pearson Correlation	.362*	.105	.399*	.224	.287	.481**	.350	.267	.657**	-.025	.313	.635**	.277	.207	.756**	.322	.163	.844**	.295	-.105	.923**	.467**	.184	.608**	.458*	.218	1	.574**	.208	.952**	
	Sig. (2-tailed)	.049	.582	.029	.234	.153	.007	.058	.154	<.001	.894	.092	<.001	.138	.273	<.001	.083	.390	<.001	.113	.582	<.001	.009	.330	<.001	.011	.246		<.001	.270	<.001	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2022/23 Sal	Pearson Correlation	.213	-.135	.111	.043	.159	.173	.142	-.073	.252	-.124	.038	.337	.061	.061	.322	.018	.083	.431*	.310	.279	.442*	.432*	.501**	.369*	.552**	.522**	.574**	1	.581**	.856**	
	Sig. (2-tailed)	.257	.478	.560	.822	.402	.361	.453	.702	.179	.513	.841	.069	.748	.749	.083	.924	.670	.017	.095	.135	.014	.017	.005	.045	.002	.003	<.001	.001		<.001	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2022/23 WR	Pearson Correlation	.173	-.395*	.034	-.136	-.157	.015	-.006	-.302	.052	-.178	-.166	.166	-.038	-.082	.084	.006	-.052	.141	.138	.261	.105	.232	.470**	.136	.277	.609**	.208	.581**	1	.238	
	Sig. (2-tailed)	.360	.031	.857	.473	.408	.936	.975	.105	.783	.346	.382	.379	.844	.666	.657	.974	.789	.457	.467	.164	.581	.218	.009	.472	.138	<.001	.270	<.001		.205	
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30
2022/23 Rev	Pearson Correlation	.342	.156	.428*	.203	.275	.506**	.376*	.279	.669**	.008	.325	.658**	.244	.201	.772**	.251	.101	.840**	.316	-.040	.912**	.387*	.211	.608**	.389*	.243	.952**	.656**	.238	1	
	Sig. (2-tailed)	.064	.409	.018	.281	.142	.004	.041	.135	<.001	.964	.079	<.001	.195	.287	<.001	.181	.601	<.001	.089	.834	<.001	.035	.262	<.001	.034	.196	<.001	<.001	.205		
	N	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	30	29	30	30	30	30	30	30	30	30	30	30	30	30

**. Correlation is significant at the 0.01 level (2-tailed).
*. Correlation is significant at the 0.05 level (2-tailed).

[1] Table 1

The table above (Table 1) shows the correlation between the three variables chosen. This represents an overview of all the season's salaries, revenues, and win rate compared to the ones of all of the other seasons. As you can see in the graphi below (table 2), we have highlighted in yellow all the strong correlations (a total of 25), and in orange all the moderate correlations (a total of 23). All the relevant correlations highlighted are positive. Between salaries and win rate, the strong correlations were in total 15, and the moderate ones 7. For the revenues and salaries, the strong correlations were in total 10, and the moderate ones 16. About the revenues and the win rate, we couldn't find any significant correlation in any of the seasons despite the relevance of this for our first hypothesis.

	Salaries		**Win Rate**		**Revenue**	
	Yellow	Orange	Yellow	Orange	Yellow	Orange
Salaries	x	x	15	7	10	16
Win Rate	15	7	x	x	0	0
Revenue	10	16	0	0	x	x

2

[2] Table 2

Season 2013/2014

Table 3 represents the correlations of salaries, revenues and win rate of 2013/2014. The salaries are positively correlated to the revenues (.566**) and the relation is strong with a significance of .001. All the other variables, when compared, are not significantly correlated, with a significance always above .005.

Correlation 2013/14

		2013/14 Sal	2013/14 WR	2013/14 Rev
2013/14 Sal	Pearson Correlation	1	.306	.566**
	Sig. (2-tailed)		.100	.001
	N	30	30	30
2013/14 WR	Pearson Correlation	.306	1	.138
	Sig. (2-tailed)	.100		.467
	N	30	30	30
2013/14 Rev	Pearson Correlation	.566**	.138	1
	Sig. (2-tailed)	.001	.467	
	N	30	30	30

**. Correlation is significant at the 0.01 level (2-tailed).

Table 3

Season 2014/2015

Table 4 represents the correlations of salaries, revenues and win rate of 2014/2015. The salaries are positively correlated to the revenues (.490**) and the relation seems to appear strong with a significance of .006. All the other variables, when compared, are not significantly correlated, with a significance always above .005.

Correlation 2014/15

		2014/15 Sal	2014/15 WR	2014/15 Rev
2014/15 Sal	Pearson Correlation	1	.255	.490**
	Sig. (2-tailed)		.174	.006
	N	30	30	30
2014/15 WR	Pearson Correlation	.255	1	-.107
	Sig. (2-tailed)	.174		.572
	N	30	30	30
2014/15 Rev	Pearson Correlation	.490**	-.107	1
	Sig. (2-tailed)	.006	.572	
	N	30	30	30

**. Correlation is significant at the 0.01 level (2-tailed).

Table 4

Season 2015/2016

Table 5 represents the correlations of salaries, revenues and win rate of 2015/2016. The salaries are positively correlated to the win rate (.562**) and the relation is strong with a significance of .001. All the other variables, when compared, are not significantly correlated, with a significance always above .005.

Correlation 2015/16

		2015/16 Sal	2015/16 WR	2015/16 Rev
2015/16 Sal	Pearson Correlation	1	.562**	.327
	Sig. (2-tailed)		.001	.078
	N	30	30	30
2015/16 WR	Pearson Correlation	.562**	1	.092
	Sig. (2-tailed)	.001		.628
	N	30	30	30
2015/16 Rev	Pearson Correlation	.327	.092	1
	Sig. (2-tailed)	.078	.628	
	N	30	30	30

**. Correlation is significant at the 0.01 level (2-tailed).

Table 5

Season 2016/2017

Table 6 represents the correlations of salaries, revenues, and win rate of 2016/2017. All the other variables in this specific season, when compared, are not significantly correlated, with a significance always above .005.

Correlation 2016/17

		2016/17 Sal	2016/17 WR	2016/17 Rev
2016/17 Sal	Pearson Correlation	1	.342	.022
	Sig. (2-tailed)		.065	.908
	N	30	30	30
2016/17 WR	Pearson Correlation	.342	1	.177
	Sig. (2-tailed)	.065		.348
	N	30	30	30
2016/17 Rev	Pearson Correlation	.022	.177	1
	Sig. (2-tailed)	.908	.348	
	N	30		30

Table 6

Season 2017/2018

Table 7 represents the correlations of salaries, revenues and win rate of 2017/2018. The salaries are positively correlated to the win rate (.636**) and the relation is strong with a significance below .001. All the other variables, when compared, are not significantly correlated, with a significance always above .005.

Correlation 2017/18

		2017/18 Sal	2017/18 WR	2017/18 Rev
2017/18 Sal	Pearson Correlation	1	.636**	.165
	Sig. (2-tailed)		<.001	.385
	N	30	30	30
2017/18 WR	Pearson Correlation	.636**	1	.169
	Sig. (2-tailed)	<.001		.372
	N	30	30	30
2017/18 Rev	Pearson Correlation	.165	.169	1
	Sig. (2-tailed)	.385	.372	
	N	30	30	30

**. Correlation is significant at the 0.01 level (2-tailed).

Table 7

Season 2018/2019

Table 8 represents the correlations of salaries, revenues and win rate of 2018/2019. The salaries are positively correlated to the win rate (.402*) and the relation is moderate with a significance of .001. All the other variables, when compared, are not significantly correlated, with a significance always above .005.

Correlation 2018/19

		2018/19 Sal	2018/19 WR	2018/19 Rev
2018/19 Sal	Pearson Correlation	1	.402*	.325
	Sig. (2-tailed)		.030	.080
	N	30	30	30
2018/19 WR	Pearson Correlation	.402*	1	.121
	Sig. (2-tailed)	.030		.533
	N	30	30	30
2018/19 Rev	Pearson Correlation	.325	.121	1
	Sig. (2-tailed)	.080	.533	
	N	30	30	30

*. Correlation is significant at the 0.05 level (2-tailed).

Table 8

Season 2019/2020

Table 9 represents the correlations of salaries, revenues and win rate of 2019/2020. The salaries are positively correlated to the win rate (.392*) and the relation is moderate with a significance of .001. All the other variables, when compared, are not significantly correlated, with a significance always above .005.

Correlation 2019/20

		2019/20 Sal	2019/20 WR	2019/20 Rev
2019/20 Sal	Pearson Correlation	1	.392*	.267
	Sig. (2-tailed)		.032	.154
	N	30	30	30
2019/20 WR	Pearson Correlation	.392*	1	-.083
	Sig. (2-tailed)	.032		.663
	N	30	30	30
2019/20 Rev	Pearson Correlation	.267	-.083	1
	Sig. (2-tailed)	.154	.663	
	N	30	30	30

*. Correlation is significant at the 0.05 level (2-tailed).

Table 9

Season 2020/2021

Table 10 represents the correlations of salaries, revenues and win rate of 2020/2021. The salaries are positively correlated to the win rate (.445*) and the relation is moderate with a significance of .001. All the other variables, when compared, are not significantly correlated, with a significance always above .005.

Correlation 2020/21

		2020/21 Sal	2020/21 WR	2020/21 Rev
2020/21 Sal	Pearson Correlation	1	.445*	.051
	Sig. (2-tailed)	.014	.791	
	N	30	30	30
2020/21 WR	Pearson Correlation	.445*	1	.248
	Sig. (2-tailed)	.014		.186
	N	30	30	30
2020/21 Rev	Pearson Correlation	.051	.248	1
	Sig. (2-tailed)	.791	.186	
	N	30	30	30

*. Correlation is significant at the 0.05 level (2-tailed).

Table 10

Season 2021/2022

Table 11 represents the correlations of salaries, revenues and win rate of 2021/2022. The salaries are positively correlated to revenues (.458*) and the relation is moderate with a significance of .001. All the other variables, when compared, are not significantly correlated, with a significance always above .005.

Correlation 2021/22

		2021/22 Sal	2021/22 WR	2021/22 Rev
2021/22 Sal	Pearson Correlation	1	.349	.458*
	Sig. (2-tailed)		.059	.011
	N	30	30	30
2021/22 WR	Pearson Correlation	.349	1	.218
	Sig. (2-tailed)	.059		.246
	N	30	30	30
2021/22 Rev	Pearson Correlation	.458*	.218	1
	Sig. (2-tailed)	.011	.246	
	N	30	30	30

*. Correlation is significant at the 0.05 level (2-tailed).

Table 11

Season 2022/2023

Table 12 represents the correlations of salaries, revenues and win rate of 2022/2023. The salaries are positively correlated to the win rate (.581**) and the relation is strong with a significance below .001. Same situation is observed between the salaries and revenues with a strong positive correlation of .656**. All the other variables, when compared, are not significantly correlated, with a significance always above .005.

Correlation 2022/23

		2022/23 Sal	2022/23 WR	2022/23 Rev
2022/23 Sal	Pearson Correlation	1	.581**	.656**
	Sig. (2-tailed)		<.001	<.001
	N	30	30	30
2022/23 WR	Pearson Correlation	.581**	1	.238
	Sig. (2-tailed)		<.001	.205
	N	30	30	30
2022/23 Rev	Pearson Correlation	.656**	.238	1
	Sig. (2-tailed)	<.001	.205	
	N	30	30	30

**. Correlation is significant at the 0.01 level (2-tailed).

Table 12

Conclusions

After analysing the correlation between the three variables chosen in each of the 10 seasons, we can conclude that generally there is a strong positive correlation between salaries and win rate and between salaries and revenues, while none between win rate and revenues. Our outcomes are based on the following observation: in six out of the ten seasons analysed, there was a positive correlation between salaries and win rate and in four out of the ten seasons analysed, there was a positive correlation between salaries and revenues.

In just one season (2016/2017) there wasn't any correlation between all three variables.

Although we can't state any type of causation, it's possible assuming that these findings are consistent with our idea that teams with higher salaries tend to have higher chances to win more games and generate greater revenue.

Relevant teams' analysis

MILWAUKEE BUCKS

We identified the case of Milwaukee Bucks as very interesting to analyze (Table 13). The team's win rate and revenue are strongly and positively correlated with salaries, with a value of .768 (for win rate and salaries) and .942 (for revenues and salaries) and a respective significance of .009 and below .001. The most interesting data is the relation between win rate and revenues: in the analysis of the seasons, this relation was never found. The correlation is still positive and moderate (.758) with a *p*-value of .011. Los Angeles Clippers and Denver Nuggets had an output similar to this one.

Correlation Milwaukee Bucks

		Salaries	Win Rate	Revenues
Salaries	Pearson Correlation	1	.768**	.942**
	Sig. (2-tailed)		.009	<.001
	N	10	10	10
Win Rate	Pearson Correlation	.768**	1	.758*
	Sig. (2-tailed)	.009		.011
	N	10	10	10
Revenues	Pearson Correlation	.942**	.758*	1
	Sig. (2-tailed)	<.001	.011	
	N	10	10	10

**. Correlation is significant at the 0.01 level (2-tailed).

*. Correlation is significant at the 0.05 level (2-tailed).

Table 13

PHILADELPHIA 76ers

Philadelphia 76ers is a very similar case to the one presented above. The team's win rate and revenue are strongly and positively correlated with salaries, with a value of .909 (for win rate and salaries) and .895 (for revenues and salaries) and a significance below .001 for both of the relations (Table 14). The most interesting data is, again, the relation between win rate and

revenues: in the analysis of the seasons, this relation was never found. The correlation in this case is positive and strong (.880) with a *p* value below .001.

Correlation Philadelphia 76ers

		Salaries	Revenues	Win Rate
Salaries	Pearson Correlation	1	.895**	.909**
	Sig. (2-tailed)		<.001	<.001
	N	10	10	10
Revenues	Pearson Correlation	.895**	1	.880**
	Sig. (2-tailed)	<.001		<.001
	N	10	10	10
Win Rate	Pearson Correlation	.909**	.880**	1
	Sig. (2-tailed)	<.001	<.001	
	N	10	10	10

**. Correlation is significant at the 0.01 level (2-tailed).

Table 14

CHARLOTTE HORNETS

Charlotte Hornets was another case worth analyzing. The team presents no significant correlations between any of our values. This data will surely help us in the final analysis of the null hypothesis. Same output was found when analysisng the correlation table of the Los Angeles Lakers.

Correlation Charlotte Hornets

		Salaries	Win Rate	Revenues
Salaries	Pearson Correlation	1	-.305	.567
	Sig. (2-tailed)		.392	.087
	N	10	10	10
Win Rate	Pearson Correlation	-.305	1	-.177
	Sig. (2-tailed)	.392		.625
	N	10	10	10
Revenues	Pearson Correlation	.567	-.177	1
	Sig. (2-tailed)	.087	.625	
	N	10	10	10

Table 15

SAN ANTONIO SPURS

San Antonio Spurs was another case that we decided to analyze further (Table 16). The reason was for the moderate negative relation that we found between win rate and revenues (-.699) with a significance of .025. This means that for this team, the values of the variables were inversely proportional. Another interesting data is the moderate positive relation between revenues and salaries (.694) with a *p-value* of .025.

Correlation San Antonio Spurs

		Salaries	Win Rate	Revenues
Salaries	Pearson Correlation	1	-.587	.694*
	Sig. (2-tailed)		.074	.026
	N	10	10	10
Win Rate	Pearson Correlation	-.587	1	-.699*
	Sig. (2-tailed)	.074		.025
	N	10	10	10
Revenues	Pearson Correlation	.694*	-.699*	1
	Sig. (2-tailed)	.026	.025	
	N	10	10	10

* Correlation is significant at the 0.05 level (2-tailed).

Table 16

Conclusions

After analyzing the cases of the teams, it's important to point out some interesting aspects. Through the studies conducted, correlations were identified 4 times (with the Milwaukee Bucks, Denver Nuggets, Philadephia 76ers, and Los Angeles Clippers) between all three variables. Surprisingly correlations between win rate and revenue were found, which had not emerged in the seasonal studies.

The aforementioned correlation was identified a total of 8 times, including 2 strong and 6 moderate, while that between win rate and salaries only 4 times, a small number compared with the 22 of the seasonal studies.

The correlation between revenues and salaries is persistent as for the seasons, also in the case of teams. Specifically, out of 30 teams analyzed, there were 27 significant correlations between revenues and salaries of which 22 were strong and 5 were moderate.

Another important finding is the case of the Los Angeles Clippers in which the correlation was not only significant in all three variables but negative between revenues and win rate and between win rate and salaries. Analyzing the LAC data, we can therefore hypothesize that the higher the revenues and salaries the lower the team's win rates.

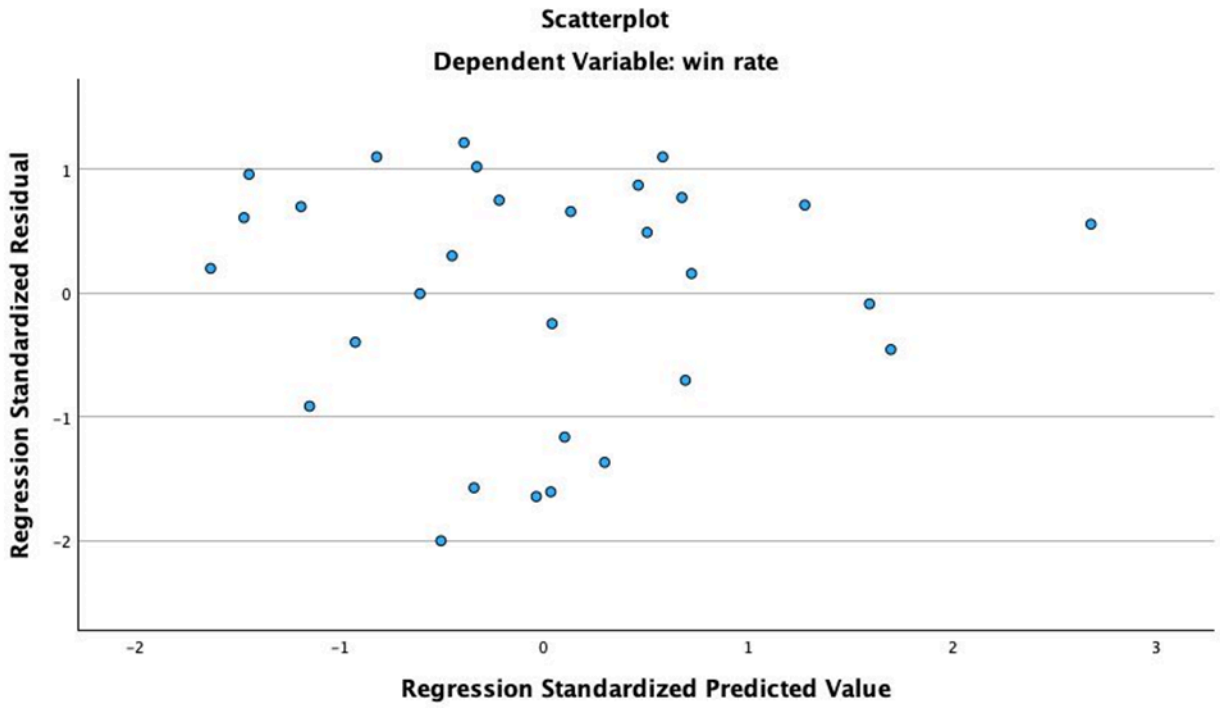
In conclusion, it is important to emphasize how through different data sets it is possible to study the phenomenon and derive such different insights from it, and at the same time better understand the context to guide a deeper analysis of the solutions.

Regression Model

Following the 5 presuppositions of the Regression Model, we have concluded that our regression model is, indeed, confirmed.

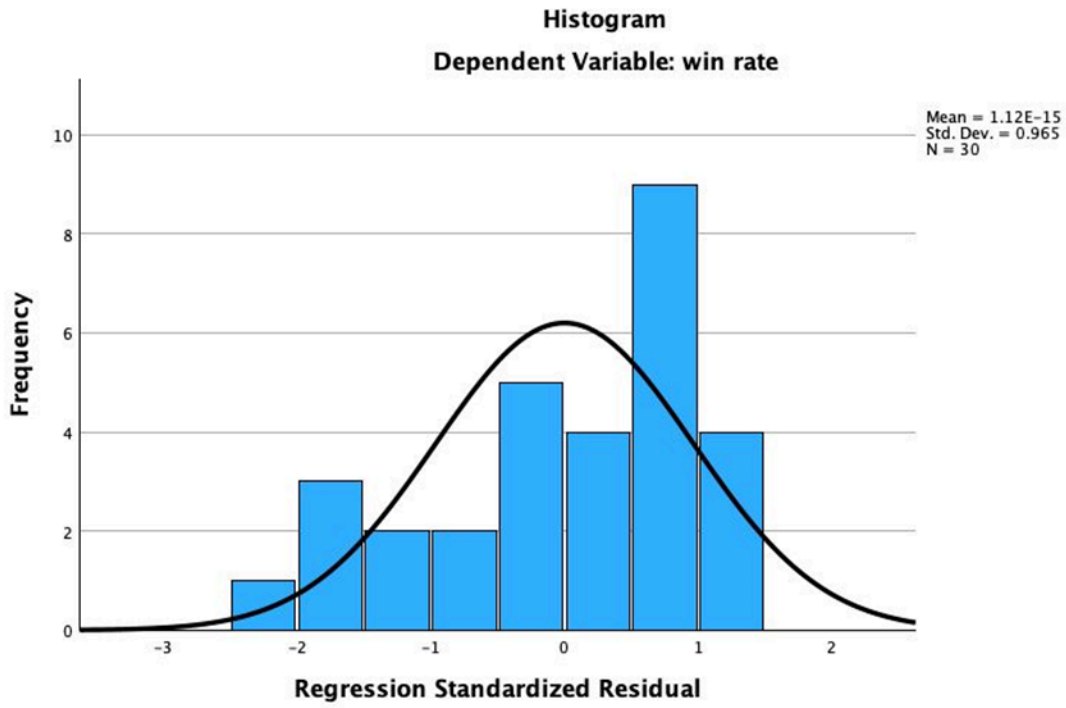
EXISTENCE OF HOMOSCEDASTICITY

Firstly, we must analyse our scatterplot graphic to confirm its homoscedasticity which is, in fact, confirmed, since it does not follow any tendency/trend



RESIDUALS MUST HAVE A NORMAL DISTRIBUTION

To confirm this one, we must analyse the Histogram, and confirm that the curve is bell-shaped, which is also true in our analysis.



Inexistence of outlier data

By analysing our residual statistics, we must consider the Std. Predicted values (Min and Max) and the Std. Residual values (Min and Max). The minimum values should not go under the -3 value, nor should the maximum values exceed 3.

Residuals Statistics[a]

	Minimum	Maximum	Mean	Std. Deviation	N
Predicted Value	45.4163%	62.3516%	51.8267%	3.93089%	30
Residual	-16.89820%	10.25881%	0.00000%	8.15148%	30
Std. Predicted Value	-1.631	2.677	.000	1.000	30
Std. Residual	-2.000	1.214	.000	.965	30

a. Dependent Variable: win rate

With this, we confirm the normality of our distribution, and don't identify any outlier values.

ABSENCE OF MULTICOLLINEARITY

To confirm the absence of multicollinearity, we must analyse the VIF values and the Tolerance values.

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients			95.0% Confidence Interval for B		Collinearity Statistics	
		B	Std. Error	Beta	t	Sig.	Lower Bound	Upper Bound	Tolerance	VIF
1	(Constant)	7.488	17.764		.422	.677	-28.960	43.937		
	Salaries	3.358E-7	.000	.394	2.268	.032	.000	.000	.996	1.004
	Revenues	2.648E-8	.000	.161	.924	.363	.000	.000	.996	1.004

a. Dependent Variable: win rate

The first, to deny the existence of multicollinearity must be under the value of 10, while the Tolerance value should be higher than 0,1, which we also confirm.

Residual Independence

Lastly, to analyse the independence of the residual values, the Durbin-Watson value should remain between 0 and 4, and, desirably, the closest to 2 as possible.

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin–Watson
1	.434[a]	.189	.129	8.44800%	1.190

a. Predictors: (Constant), Revenues, Salaries

b. Dependent Variable: win rate

Our model, with a value of 1.190 indicates a positive autocorrelation, while if it was higher than two, it would suggest a negative autocorrelation.

While still analysing our Model Summary table, the R value confirms that our model will be correct in 43.4% of the cases, for the next 10 years – the same timespan we analysed.

Conclusions

Summary of Findings

After this thorough analysis, we now want to summarise our results by answering our hypotheses as well as our initial research question:

H0: Revenues of teams, salaries of teams and win rate are not associated.
While the null hypothesis was initially proposed, we deny it with exceptions observed in only three instances. Notably, the teams Charlotte Hornets and the Los Angeles Lakers, along with the entire NBA season 2016/2017, displayed no significant correlation among the three variables.

H1: Revenues of teams are associated with the teams' win rate.
We deny our initial hypothesis, which predicted a relationship between team revenue and win rate, with a few exceptions, as we only found a few cases where this was correct

H2: Team' salaries are associated with the teams' win rate.
Our examination of this hypothesis found consistent correlations in both data sets, supporting the idea that team salaries are related to teams' win rates and confirming this hypothesis.

H3: Revenues of teams are associated with the teams' salaries.
Similar to H2, our findings indicate a repeated correlation between team revenues and team salaries in both datasets, which also confirms this hypothesis.

How does the luxury tax create an unfair competitive system in the NBA?
Regarding our initial research question, one can therefore conclude that while increased financial resources do not guarantee success, we have found that two of the four teams with the highest correlations between variables - have won the NBA Championship in recent years, the Milwaukee Bucks in 2021 and the Denver Nuggets in 2023. This raises questions about the impact of the luxury tax on creating an unfair competition system.

The strong correlation between salaries and revenues as well as salaries and win rate prohibits us from claiming causality. However, based on the results, it is likely that as a team's revenue increases, so will their salaries, potentially increasing their chances of success.

Limitations

The concerns of our research are connected to the fact that following elements could influence the outcome of the games:

- Home advantage, as explained by Josh Leota et al, (2022), increases the win probability.
- Load management is when a healthy player is held out of playing in a game for rest or precautionary reasons, this can change the fate of the game as well.
- Tanking is the art of creating a purposefully bad team with the intention of losing games to gain high draft picks. Tanking aims to gain this player with top draft pick, and to ultimately win a championship with the core constructed while tanking.
- Luck influences the winning rate because even the greatest players have some bad days and don't perform well, but even when they do, simple plays can influence the outcome of the game.
- One big aspect of luck is players' injuries. If they are unlucky, they can be even out for the rest of the season and especially when the stars of the teams get injured it can influence the winning rate.

The limitations of our research don't stop here: for example we used resources from the past season, where the luxury tax system was different from now. Additionally, some of the resources

were taken from research on other types of sports, so we don't know if they can be completely comparable.

Because we neither have the capacity, the time, nor the necessary connections, we cannot conduct a qualitative analysis. If we had the opportunity, we would do it as follows:

Indirect observation

Why are we using it?

To understand the perception people have of luxury tax. It's a broader audience that we can get access to and because they are anonymous, their opinions will most likely be more honest. In addition, there are people very interested in the topic and very keen on analysing it in depth and with real-time reactions.

How can it help with our research question?

This can be important for our research questions because we can analyse the digital world that surrounds the NBA as well as the sentiment of the public with a particular focus on the luxury tax. This can allow us to get insight from the public, a part of the business that is paramount nowadays.

Furthermore, a lot of fans produce a huge amount of UGC (user-generated content) which goes much deeper into the situations that are happening in the NBA, and most of the time, even huge broadcast companies like ESPN report about it.

There are also a lot of communities on various social media platforms that discuss relevant topics surrounding the NBA.

What steps are we following?

1. Search different platforms on the internet for opinions on our topic: e.g., Twitter, Reddit, Instagram pages, blogs, etc...
2. Analyse them.

In this part we will compare the general feeling of people between different social media channels, taking into consideration the environment of the platform and the context of each comment.

Focus Groups

Through the insights of the indirect observation, we can create two different types of focus groups, one with fans and one with NBA professionals, such as players, coaches, owners and other representatives of the league, to speak about the topic “luxury tax”.

We will give each group the same sets of questions. We want to do it to have a different type of approach to the study because we think that through a real confrontation (that is not anonymous like in the case of indirect observation) we can find a lot of insights.

Additionally, we would like to use the information we gained for the creation of our questions that specialists can then analyse and validate.

Question analysis

We would take a group of specialists in the field and in statistics that will validate our questionnaire, to use them for our in-depth interviews.

In-depth interview

Why are we using that?

So that we can gather different opinions on the luxury tax from people of different ages and experiences, leading to an interview with a structured selection of questions. This will let us add more insights about not only how the luxury tax is perceived but also new solutions that could be used to solve the current situation.

How can it help with our research question?

It can give us a deeper insight into people's opinions and an overview, without the anonymity factor, which can possibly lead to more honest and in-depth answers. Thanks to this approach, we can also find possible solutions and discuss them with the involved elements.

What steps are we following?

1. **Define the questions.**

A draft for the questions selected:

1.1. What do you understand under the term "luxury tax" in the context of the NBA?
1.2. What is your opinion about the luxury tax?
1.3. Do you think it's fair?
1.4. Do you think with the luxury tax every team has an equal chance to win the championship?
1.5. Do you have an idea how to improve the luxury tax system to make it fairer?
1.6. Do you think the NBA should introduce a "hard" salary cap instead of the current system with a "soft" salary cap and a luxury tax?

(Instruction for the interviewer) Describe the problem more in-depth by using the example of the Golden State Warriors.

1.7. What do you think could be an alternative way to solve the problem we described?

2. **Find interview partners.**

Our sample will be composed of a group of 5-6 people who already know how the luxury tax works and understand the NBA itself. No ranges of age will be used either of experience.

3. **Conduct in-depth interviews.**
4. **Analyse the results.**

Future Applications and Recommendations

Our ongoing project, centred on the analysis of NBA teams' data of 2022/2023 season, presents promising applications and strategic recommendations for stakeholders across the sports industry.

First, the insights from our meticulous analysis aim to revolutionise strategic decision-making for NBA teams. By unravelling the complex relationship between team revenues, player salaries, and win rates, our findings provide teams with a roadmap to optimise resource allocation and enhance overall fair competitiveness.

Beyond the team level, our project has implications for league management and policy formulation. The NBA and other sports leagues can use our data-driven insights to refine salary cap rules, luxury tax structures and revenue sharing mechanisms. This approach ensures a delicate balance between promoting competitive parity and ensuring the financial sustainability of the league.

The implications of this work extend to player negotiations. Agents and players, with a clear understanding of the correlation between team success and financial metrics, can enter contract negotiations better informed and empowered to secure agreements reflective of their on-court contributions.

External stakeholders, such as investors and sponsors, gain a unique perspective into the financial dynamics of NBA teams. Recognising the interdependence of team success and financial metrics enables the identification of profitable investment and sponsorship opportunities within the dynamic sports industry landscape.

Importantly, our recommendations underscore the need for continuous monitoring and adaptation. As the sports landscape evolves, teams and stakeholders must remain on track, regularly updating strategies with new data to ensure relevance and effectiveness in an ever-changing environment.

After all these considerations, we came up with two solutions that could assure a fairer competitive system in the NBA.

The first one would be easier to apply, namely erase the luxury tax and adhere instead to a hard cap like the one applied by the NFL, so a threshold that must not be exceeded by even one dollar. However, this must be done while maintaining the current minimum threshold, in a way that each team is going to spend the same amount of money and a factor like the revenue would not be influent anymore on the win rate.

While the second and more articulated idea that would require more discussion with the teams is to tailor the luxury tax rates to a team's revenue, considering both local and league-wide earnings. Teams with higher revenues would face a more stringent luxury tax, creating a direct link between financial success and the financial obligations imposed by exceeding the salary cap. This approach aligns with the principle that teams generating more revenue should contribute more to maintaining competitive balance and league sustainability.

These proposed solutions aim to enhance the effectiveness and fairness of the luxury cap system by incorporating progressive elements, incentives, and revenue considerations. The goal is to strike a balance that encourages competitive spending, discourages excessive frugality, and ensures that the burden of the luxury tax is commensurate with a team's financial capacity and success.

These proposed solutions aim to enhance the effectiveness and fairness of the luxury cap system. The goal is to strike a balance that encourages competitive spending, while discouraging excessive frugality and ensuring that the burden of the luxury tax is commensurate with a team's financial capacity and success.

In conclusion, our project transcends statistical analysis, offering actionable insights that promise to shape a more competitive, sustainable, and engaging future for the NBA and the broader sports industry. The collaborative application of this knowledge stands as a testament to its transformative potential.

Appendices

References

Hypothesis H1:

Ozanian, M. (2021, October 18). The NBA's most valuable teams 2021-22: New York Knicks lead a trio now worth over $5 billion each. Forbes. https://www.forbes.com/sites/mikeozanian/2021/10/18/the-nbas-most-valuable-teams-2021-22-new-york-knicks-lead-a-trio-now-worth-over-5-billion-each/?sh=1948efb7d32c

Hypothesys 2:

Deeks, M. (2022, July 2). A complete history of NBA luxury tax payments, 2001-2022. Forbes. https://www.forbes.com/sites/markdeeks/2022/07/01/a-complete-history-of-nba-luxury-tax-payments-20012022/

Garner, B. R. (n.d.). Analyzing the impact of salary discrepancies between professional sports organizations. Scholar Commons. https://scholarcommons.sc.edu/senior_theses/618/

Toporek, B. (2022, August 2). Sorry, Golden State Warriors, but the NBA's luxury tax is working as intended. *Forbes*. https://www.forbes.com/sites/bryantoporek/2022/08/02/sorry-golden-state-warriors-but-the-nbas-luxury-tax-is-working-as-intended/

Toporek, B. (2022b, September 21). How NBA expansion could impact collective bargaining agreement negotiations. *Forbes*. https://www.forbes.com/sites/bryantoporek/2022/09/21/how-nba-expansion-could-impact-collective-bargaining-agreement-negotiations/?sh=615beab83f5

Ribeiro, D. (17, April 2018). *Verificando os pressupostos de uma regressão. (Utilizando o SPSS)*. Linkedin. Retrieved January 16, 2024, from

https://www.linkedin.com/pulse/verificando-os-pressupostos-de-uma-regress%C3%A3o-o-spss-ribeiro/?originalSubdomain=pt

Hypothesis 3:

Menon, P. (n.d.). REDEF ORIGINAL: Pay To Play: Why It's time to end the NBA's luxury tax. REDEF. https://redef.com/original/pay-to-play-why-its-time-to-end-the-nbas-luxury-tax

Bradbury JC. Determinants of revenue in sports leagues: An empirical assessment. Economic Inquiry. 2019. Jan;57(1):121–4

These are the 2022/23 salaries of all NBA teams. (n.d.). HoopsHype. https://hoopshype.com/salaries/2022-2023/

Data Sources:

Limitations:

Gerrard, B. (2021, December 8). What are the odds: A statistical analysis of tanking in the NBA. Medium. https://towardsdatascience.com/what-are-the-odds-a-statistical-analysis-of-tanking-in-the-nba-2c5fe228cd67

Road, R. (2023, July 17). What is NBA Load Management? https://www.rookieroad.com/basketball/what-is-nba-load-management-4237574/

Leota, J., Hoffman, D., Mascaro, L., Czeisler, M. É., Nash, K., Drummond, S., Anderson, C., Rajaratnam, S. M. W., & Facer-Childs, E. R. (2022). Home is where the hustle is: the influence of crowds on effort and home advantage in the National Basketball Association. *Journal of Sports Sciences*, *40*(20), 2343–2352. https://doi.org/10.1080/02640414.2022.2154933